

Prominent theories of emotion have long held that alterations in one's internal physiological state are necessary for the experience of emotion. Interoception involves both the unconscious and conscious perception of one's internal physiological state. The insula (or insular cortex), located deep within the lateral fissure on both sides of the brain, appears to be largely responsible for interoception. Studies suggest that the posterior insula (PI) is involved with processing homeostatic afferent information such as pain, temperature, hunger, thirst, and the impulse to breathe. In contrast, the anterior insula (AI) appears to process information regarding emotional awareness, the conscious experience of emotion.

A recent study evaluated AI activity using functional magnetic resonance imaging (fMRI). Using fMRI, the participants' brain activity was monitored during two different tasks: an emotion-eliciting task and an interoceptive task.

Task 1

Participants watched 10 short videos of people recalling personal events that were either negative (eg, surviving a mass shooting attack) or positive (eg, seeing a loved one after a year apart). After each video, participants rated their reactions on a Likert scale from 1 (no emotional reaction) to 10 (extreme emotional reaction).

Task 2

Participants were instructed to monitor their own heart rate and press a button each time they felt their heartbeat. Participants' heart rates were externally monitored to gauge the accuracy of their responses.

Results

Neuroimaging results suggested that both tasks elicited activity in the AI. Furthermore, there was a high degree of overlap in the activity of the right AI and surrounding limbic and paralimbic brain regions at both the individual and group levels.

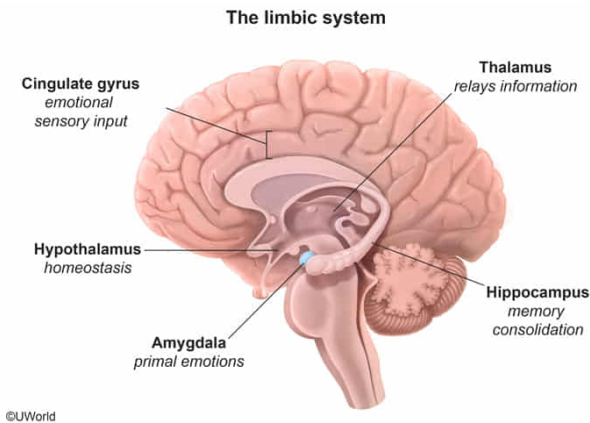
The results of the study suggest which of the following brain regions was also activated?

- ✔ ☒ A. Amygdala [86%]
- ☐ B. Basal ganglia [6%]
- ☐ C. Cerebellum [2%]
- ☐ D. Somatosensory cortex [4%]

Correct

86% answered correctly

Explanation:



According to the passage, fMRI results suggest a high degree of overlap in the right AI and surrounding limbic and paralimbic regions of the brain. The **limbic system** is a network of brain regions that are involved in **emotion, learning, and memory**. This system is composed of many structures, including the amygdala, hippocampus, hypothalamus, and cingulate gyrus.

The **amygdala** is a small, almond-shaped structure located deep within the temporal lobe on both sides of the brain. Studies have shown that neurons of the amygdala fire while processing more **primal emotions**, such as fear, anger, and sexual arousal. The amygdala also plays a role in fear conditioning and the encoding of other emotionally charged memories.

(Choice B) The basal ganglia contain a high concentration of dopamine-producing neurons, the function of which is to produce smooth, purposeful movements and inhibit excessive movement. Loss of dopamine-producing neurons in the substantia nigra of the basal ganglia results in the characteristic resting tremor seen in Parkinson disease. The basal ganglia are not part of the limbic system.

(Choice C) The cerebellum, which lies below the cerebral cortex at the back of the brain, is responsible for motor coordination and balance. The cerebellum is not part of the limbic system.

(Choice D) The somatosensory cortex is located just posterior to the lateral fissure in the parietal lobe. This region of the brain processes sensory information from touch and maps it to a specific area of the body. The somatosensory cortex is not part of the limbic system.

Educational objective:

The limbic system is involved in emotion, learning, and memory, and contains

several structures, including the amygdala, hippocampus, hypothalamus, and cingulate gyrus. The amygdala functions in the processing of primal emotions (eg, fear, anger, sexual arousal) and the encoding of other emotionally charged memories.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400405

System:
Motivation, Emotion, Attitudes, Personality, and Stress

Prominent theories of emotion have long held that alterations in one's internal physiological state are necessary for the experience of emotion. Interoception involves both the unconscious and conscious perception of one's internal physiological state. The insula (or insular cortex), located deep within the lateral fissure on both sides of the brain, appears to be largely responsible for interoception. Studies suggest that the posterior insula (PI) is involved with processing homeostatic afferent information such as pain, temperature, hunger, thirst, and the impulse to breathe. In contrast, the anterior insula (AI) appears to process information regarding emotional awareness, the conscious experience of emotion.

A recent study evaluated AI activity using functional magnetic resonance imaging (fMRI). Using fMRI, the participants' brain activity was monitored during two different tasks: an emotion-eliciting task and an interoceptive task.

Task 1

Participants watched 10 short videos of people recalling personal events that were either negative (eg, surviving a mass shooting attack) or positive (eg, seeing a loved one after a year apart). After each video, participants rated their reactions on a Likert scale from 1 (no emotional reaction) to 10 (extreme emotional reaction).

Task 2

Participants were instructed to monitor their own heart rate and press a button each time they felt their heartbeat. Participants' heart rates were externally monitored to gauge the accuracy of their responses.

Results

Neuroimaging results suggested that both tasks elicited activity in the AI. Furthermore, there was a high degree of overlap in the activity of the right AI and surrounding limbic and paralimbic brain regions at both the individual and group levels.

Which of the following accurately pairs a component of emotion with the study procedure designed to evoke that component?

- ☐ A. Behavioral; rating emotional reactions on a Likert scale [15%]
- ☒ B. Cognitive; watching videos of others describing traumatic events [38%]
- ☐ C. Affective; performing tasks while in an fMRI machine [6%]
- ☐ D. Physiological; pressing a button with each heartbeat [39%]

Correct

37% answered correctly

Explanation:

The 3 components of human emotion		
Component	Description	Examples
Cognitive	Mental processes accompanying an emotion & involved in appraisal of the situation	• Thoughts • Beliefs • Expectations • Rationalization
Behavioral	Immediate outward reaction to an emotion that is involuntary & automatic	• Smiling • Gasping
Physiological	Bodily processes that accompany an emotion	• Heart rate changes • Respiration rate changes • Sweating

The three components of emotion are **cognitive**, **behavioral**, and **physiological**.

- The **cognitive component** includes all the mental processes that accompany the emotion, such as corresponding thoughts, motivations, beliefs, expectations, and an evaluation of the context/situation.
- The **behavioral component** of emotion is described as the immediate **outward reaction** that occurs in response to an emotion. These responses are typically involuntary and automatic (eg, smiling, gasping).
- The **physiological component** of emotion includes all the **bodily processes** that accompany the emotion (eg, changes in heart rate, respiration rate, sweating).

In Task 1 of the study, participants watched videos of individuals describing personal events (both negative and positive) and then rated their reactions on a scale of 1 (no emotional reaction) to 10 (extreme emotional reaction). The videos were designed to evoke the *cognitive* component of emotion because they required the viewers to think about the events being described and likely also involved their expectations and beliefs.

(Choice A) Rating an emotional reaction on a Likert scale is a voluntary and purposeful action. Therefore, this study procedure was not designed to evoke the behavioral component of emotion, which involves largely involuntary and automatic responses, such as gasping in surprise.

(Choice C) Performing tasks while in an fMRI machine is a study procedure used to measure brain activity during the tasks, not a method employed to evoke any particular component of emotion. Furthermore, because the fMRI is used as a measurement tool, any emotion it may evoke in participants would be undesirable, as this could compromise or confound the results.

(Choice D) Task 2 required participants to press a button with each heartbeat. This task was not designed to involve any emotion whatsoever but to evoke *interoception* (physiological awareness).

Educational objective:

The three components of emotion are cognitive, behavioral, and physiological. The cognitive component includes all the mental processes that accompany emotion (eg, motivations, beliefs, expectations). The behavioral component includes immediate outward reactions (eg, gasping), and the physiological component includes bodily reactions (eg, increased heart rate).

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400406

System:
Motivation, Emotion, Attitudes, Personality, and Stress

Prominent theories of emotion have long held that alterations in one's internal physiological state are necessary for the experience of emotion. Interoception involves both the unconscious and conscious perception of one's internal physiological state. The insula (or insular cortex), located deep within the lateral fissure on both sides of the brain, appears to be largely responsible for interoception. Studies suggest that the posterior insula (PI) is involved with processing homeostatic afferent information such as pain, temperature, hunger, thirst, and the impulse to breathe. In contrast, the anterior insula (AI) appears to process information regarding emotional awareness, the conscious experience of emotion.

A recent study evaluated AI activity using functional magnetic resonance imaging (fMRI). Using fMRI, the participants' brain activity was monitored during two different tasks: an emotion-eliciting task and an interoceptive task.

Task 1

Participants watched 10 short videos of people recalling personal events that were either negative (eg, surviving a mass shooting attack) or positive (eg, seeing a loved one after a year apart). After each video, participants rated their reactions on a Likert scale from 1 (no emotional reaction) to 10 (extreme emotional reaction).

Task 2

Participants were instructed to monitor their own heart rate and press a button each time they felt their heartbeat. Participants' heart rates were externally monitored to gauge the accuracy of their responses.

Results

Neuroimaging results suggested that both tasks elicited activity in the AI. Furthermore, there was a high degree of overlap in the activity of the right AI and surrounding limbic and paralimbic brain regions at both the individual and group levels.

To test the hypothesis that emotions play an adaptive role in society, researchers added a Task 3 to assess social success (higher scores = more

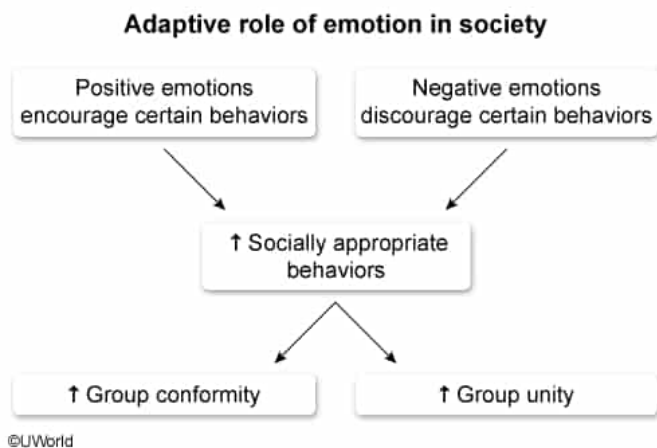
social success). This hypothesis would be best supported by:

- ☐ A. a negative correlation between Task 1 scores for negative videos and Task 3 scores. [11%]
- ✔ ☒ B. a positive correlation between Task 1 scores and Task 3 scores. [76%]
- ☐ C. a negative correlation between heart rate during Task 2 and Task 3 scores. [3%]
- ☐ D. a positive correlation between Task 2 accuracy and Task 3 scores. [9%]

Correct

77% answered correctly

Explanation:



Emotion plays an important role in individual and group behaviors. Positive emotions motivate people to engage in a behavior again whereas negative emotions cause avoidance. For example, if a child shares her toy and is praised for it, she is likely to repeat the behavior. On the other hand, if a teenager is socially shunned for wearing a new pair of shoes, he is unlikely to wear them again.

Emotions play an **adaptive role** in society by allowing humans to **emotionally connect** with others through feelings such as love, affection, and attachment, and by fostering **group unity** and **conformity**. A desire to experience certain emotions (eg, happiness, pride) and avoid others (eg, embarrassment, shame) causes people to behave in predictable, socially acceptable ways. Understanding the emotions of others and expressing emotions to others are important elements of social interaction.

A positive correlation between emotional expression scores from Task 1 (for both the positive and negative videos) and social success scores would support the hypothesis that emotions play an adaptive role in society. Individuals with stronger emotional reactions to the videos may be more understanding/perceptive of positive or negative emotions, which should correspond with an increased ability to build interpersonal connections (ie, social success).

(Choice A) The ability to have an emotional reaction to both positive and negative life experiences of others is important for social success. Therefore, there should be a positive (not negative) correlation between Task 1 scores for both negative and positive videos and Task 3 scores.

(Choice C) Because changes in heart rate reflect a variety of different emotions, both positive (eg, excitement) and negative (eg, anxiety), it is not possible to determine how heart rate might be related to social success.

(Choice D) During Task 2, participants pushed a button every time they felt their heartbeat. Task 2 accuracy would mean that button pushes corresponded with actual heart rate, suggesting that the participant demonstrated good interoceptive ability. However, it is unclear how this ability might relate to social success.

Educational objective:

Emotions play an adaptive role in society by influencing behavior and social interaction in ways that improve group unity and conformity. Individuals who are better at perceiving and understanding emotions should also have more social success.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400407

System:
Motivation, Emotion, Attitudes, Personality, and Stress

Prominent theories of emotion have long held that alterations in one's internal physiological state are necessary for the experience of emotion. Interoception involves both the unconscious and conscious perception of one's internal physiological state. The insula (or insular cortex), located deep within the lateral fissure on both sides of the brain, appears to be largely responsible for interoception. Studies suggest that the posterior insula (PI) is involved with processing homeostatic afferent information such as pain, temperature, hunger, thirst, and the impulse to breathe. In contrast, the anterior insula (AI) appears to process information regarding emotional awareness, the conscious experience of emotion.

A recent study evaluated AI activity using functional magnetic resonance imaging (fMRI). Using fMRI, the participants' brain activity was monitored during two different tasks: an emotion-eliciting task and an interoceptive task.

Task 1

Participants watched 10 short videos of people recalling personal events that were either negative (eg, surviving a mass shooting attack) or positive (eg, seeing a loved one after a year apart). After each video, participants rated their reactions on a Likert scale from 1 (no emotional reaction) to 10 (extreme emotional reaction).

Task 2

Participants were instructed to monitor their own heart rate and press a button each time they felt their heartbeat. Participants' heart rates were externally monitored to gauge the accuracy of their responses.

Results

Neuroimaging results suggested that both tasks elicited activity in the AI. Furthermore, there was a high degree of overlap in the activity of the right AI and surrounding limbic and paralimbic brain regions at both the individual and group levels.

The neuroimaging tool used for this study works by measuring which of the following?

- ☐ A. Electrical impulses [15%]
- ☐ B. Glucose metabolism [12%]
- ☐ C. Radioactive tracing [8%]
- ☒ D. Blood oxygenation [63%]

Correct

62% answered correctly

Explanation:

Neuroimaging method	Protocol	Measures
EEG	Electrodes placed on the scalp & connected to an amplifier	Electrical signals of the cortex below the skull
MEG	Helmet-like device placed over the head	Magnetic fields produced by electrical brain activity
PET	Scanner detects radioactive tracer attached to a glucose analog	Glucose metabolism in the brain
fMRI	Scanner detects the differential properties of oxyhemoglobin & deoxyhemoglobin	Blood oxygenation in the brain
EEG = electroencephalogram; fMRI = functional magnetic resonance imaging; MEG = magnetoencephalogram; PET = positron emission tomography.		

This study used **functional magnetic resonance imaging (fMRI)**, a neuroimaging procedure that relies on the principle that active neurons require increased **blood flow** for **oxygen** delivery. Neurons lack internal energy stores (ie, no glucose or oxygen), so as an active brain region depletes the local oxygen supply in the blood, freshly oxygenated blood (oxyhemoglobin) rushes to this area to replace the deoxygenated blood (deoxyhemoglobin).

The resulting change in blood flow is analyzed by detecting the differential properties of oxyhemoglobin and deoxyhemoglobin using an fMRI scanner. Researchers use this method to compare blood flow during a resting condition and an active condition. For example, the study researchers would have first assessed subjects at baseline (resting condition) and then while watching videos meant to induce emotion (active condition) and compared the regions of brain activation.

(Choice A) An electroencephalogram (EEG) measures the patterns and fluctuations of electrical impulses in the brain through electrodes placed on the scalp. This technique records brain waves, and is most often used to diagnose epilepsy, sleep disorders, and coma.

(Choices B and C) Positron emission tomography (PET) uses a radioactive tracer attached to a glucose analog to measure blood flow and glucose uptake induced by brain activity. PET is often combined with other procedures and produces a three-

dimensional image of the brain showing hypermetabolic and hypometabolic regions.

Educational objective:

Functional magnetic resonance imaging is a neuroimaging tool that measures the difference in blood oxygenation between a resting condition and an active condition. It provides an indirect measure of brain activity associated with the experimental conditions.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400408

System:
Motivation, Emotion, Attitudes, Personality, and Stress